

Way To Finance Success

MIND MAPS CFA® EXAM PRE

LEVEL 2

2016

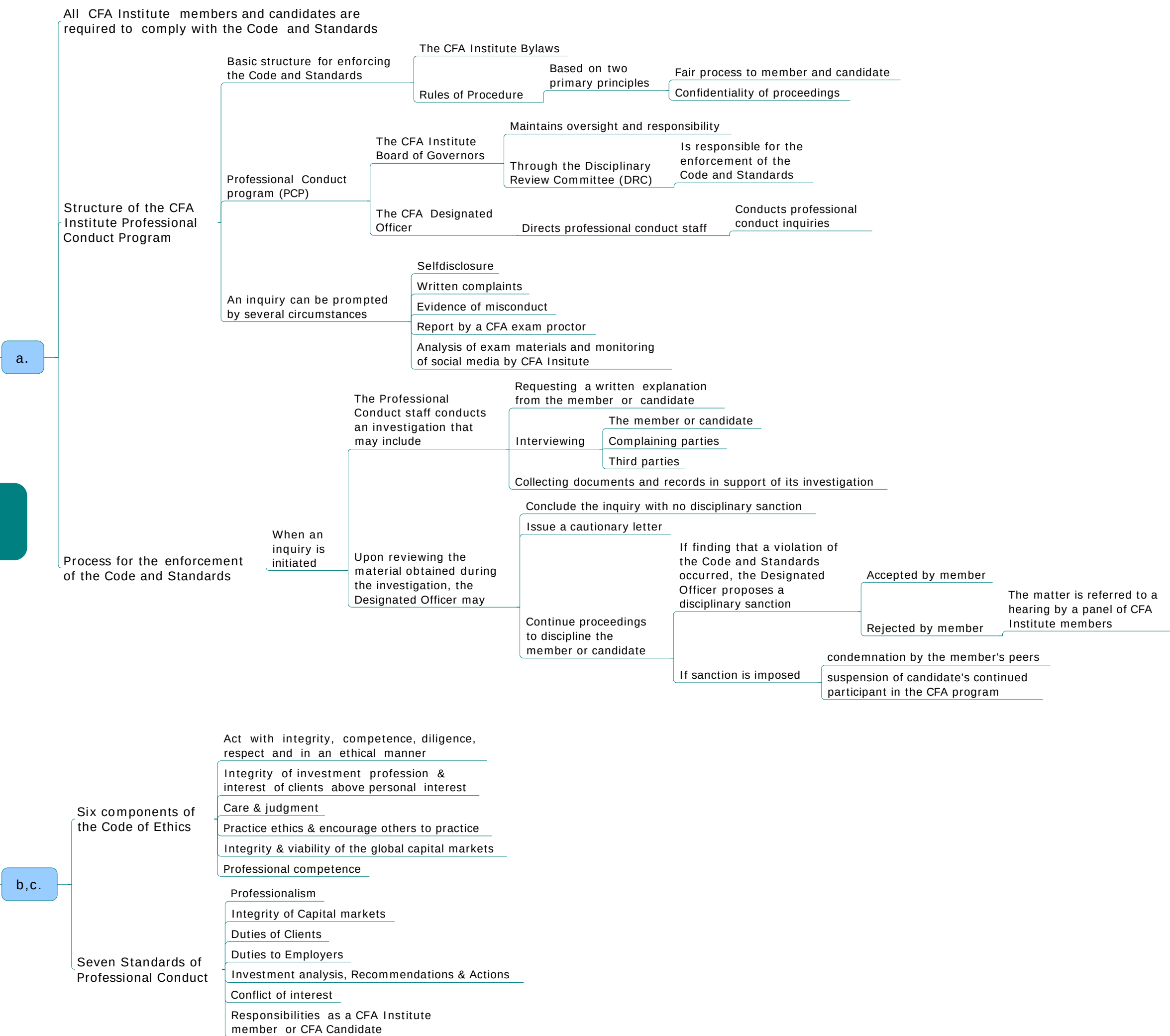
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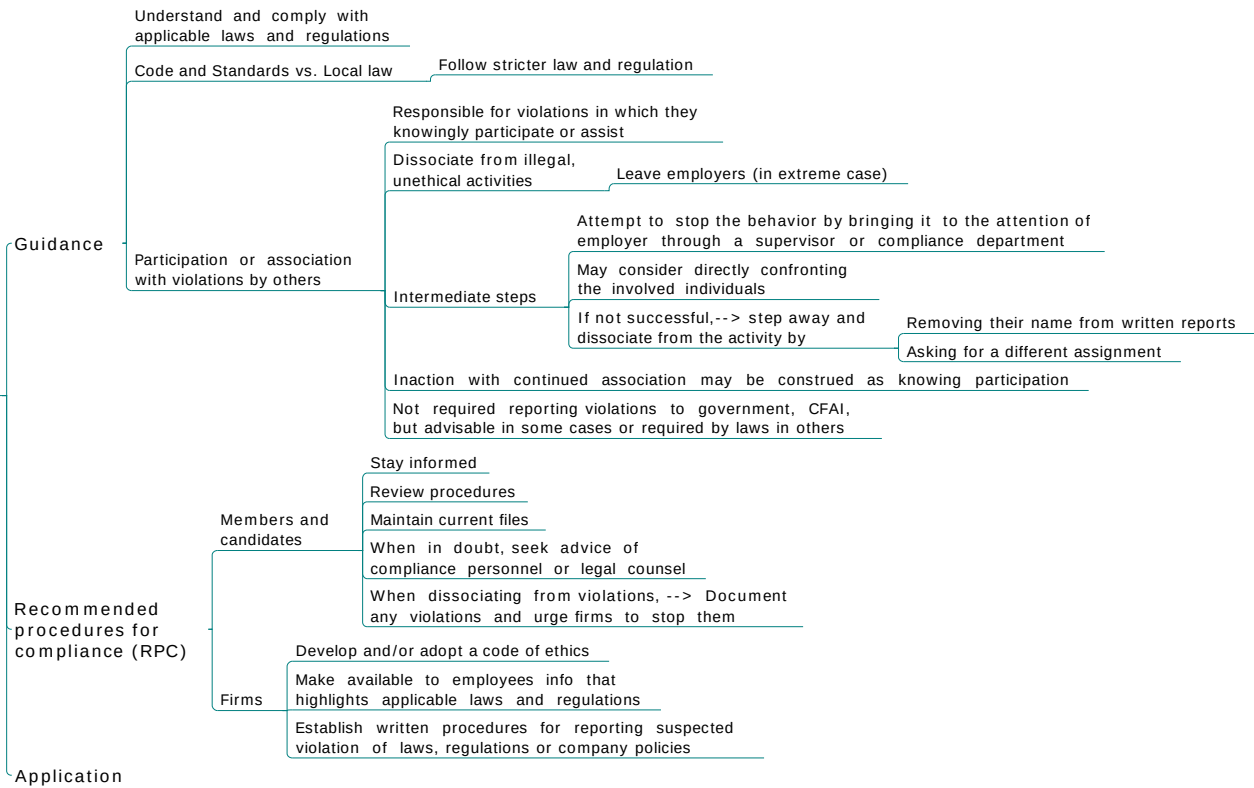
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1. Code Of Ethics And Standards Of Professional Conduct

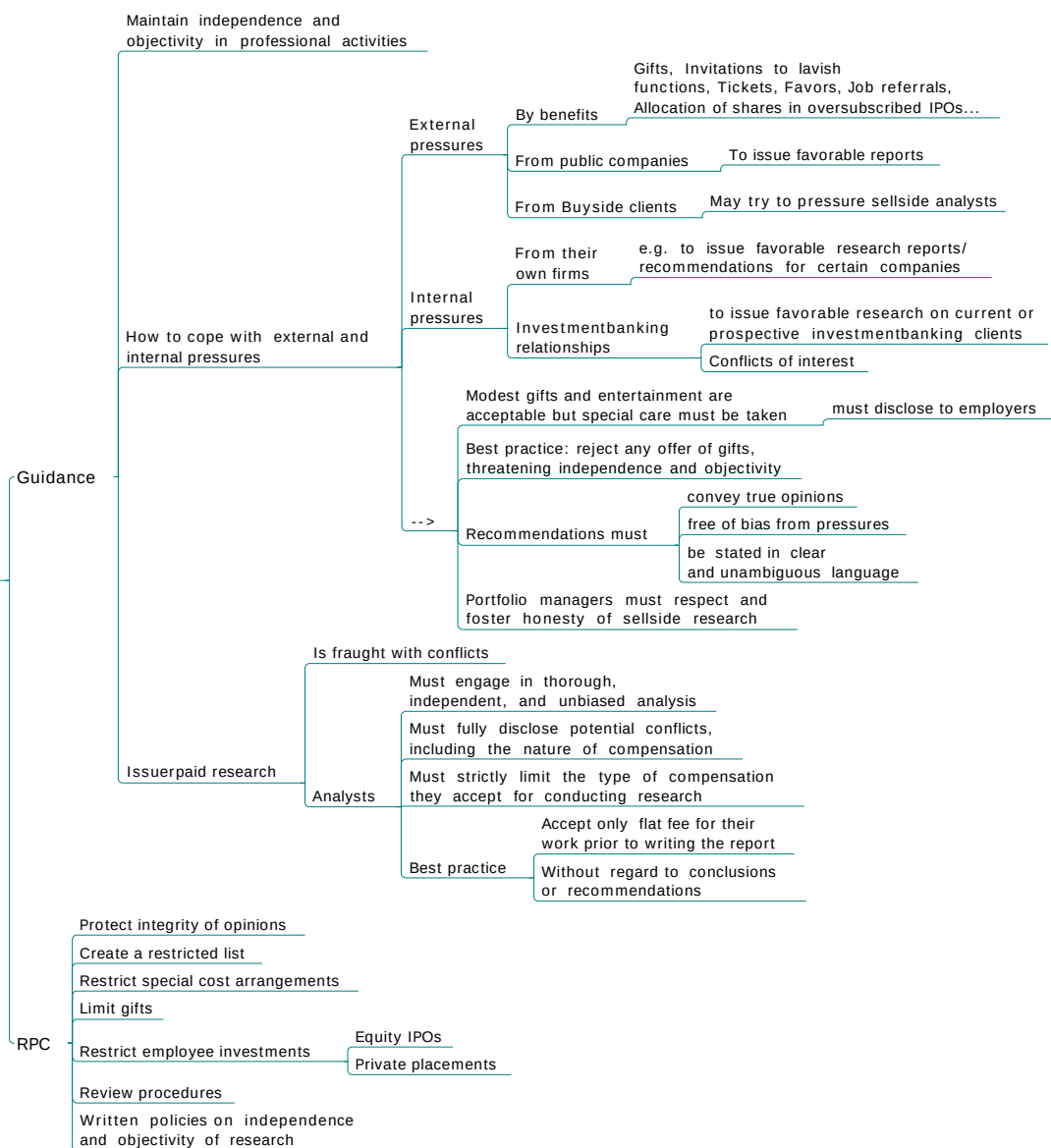


2.1 Standard I PROFESSIONALISM

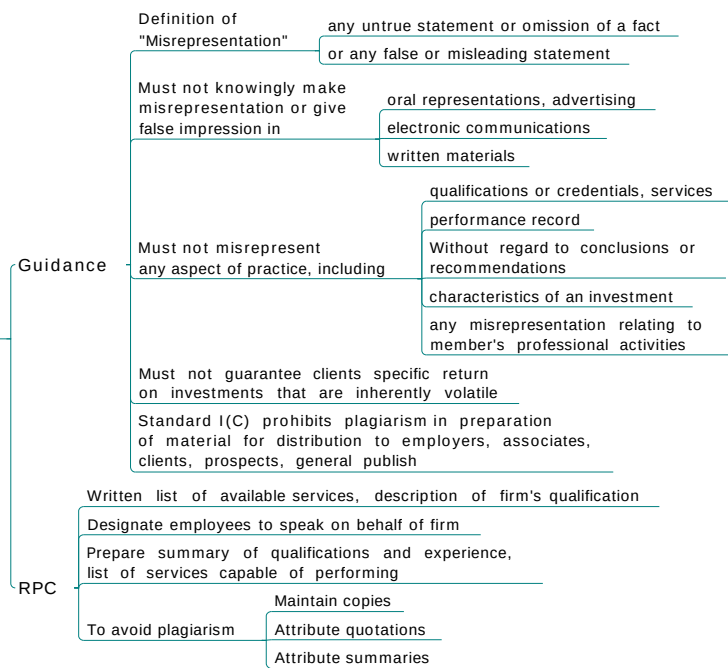
A. Knowledge of the law



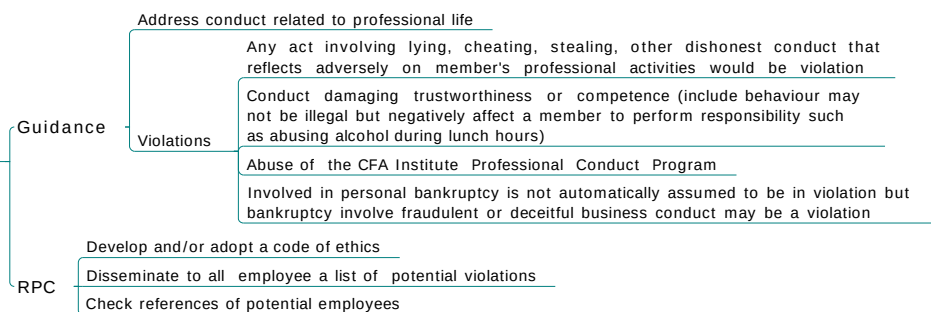
B. Independence and objectivity



C. Misrepresentation



D. Misconduct



2.2 Standard II INTEGRITY OF CAPITAL MARKETS

A. Material non-public information (MNI)

Guidance

Definition of "Material nonpublic information"

Material information

its significant impact to the price of security if it is disclosed

Reasonable investors would like to know for making decision

The reliability of the information

Non-public until

disseminated to the market place and efficient time for investors to react

Must be particularly aware of info selectively disclosed by corporations

Mosaic Theory

Analysis of Public info + nonmaterial nonpublic info --> Investment conclusion

Analysts are free to act on this collection of info without risking violation

Analysts should save and document all their research

RPC

Make reasonable efforts to achieve public dissemination of material info

If public dissemination is not possible,

Must communicate the info only to the designated supervisory and compliance personnel within the firm

Must not take investment action on the basis of the info

Must not knowingly engage in conduct inducing insiders to privately disclose MNI

Encourage firms to

adopt compliance procedures preventing misuse of MNI

develop & follow disclosure policies to ensure proper dissemination

use "firewall"

Prohibition of all proprietary trading while firm is in possession of MNI may be inappropriate

B. Market manipulation

Definition

Distort prices or artificially inflate trading volume with the intent to mislead market participants

can be related to

transactions that deceive market participants

Transactions that artificially distort prices or volume

Securing a controlling, dominant position in a financial instrument to exploit and manipulate price of a related derivative/or underlying asset

dissemination of false or misleading info

including spreading false rumors to induce trading by others

Standard II(B) not meant to

prohibit legitimate trading strategies

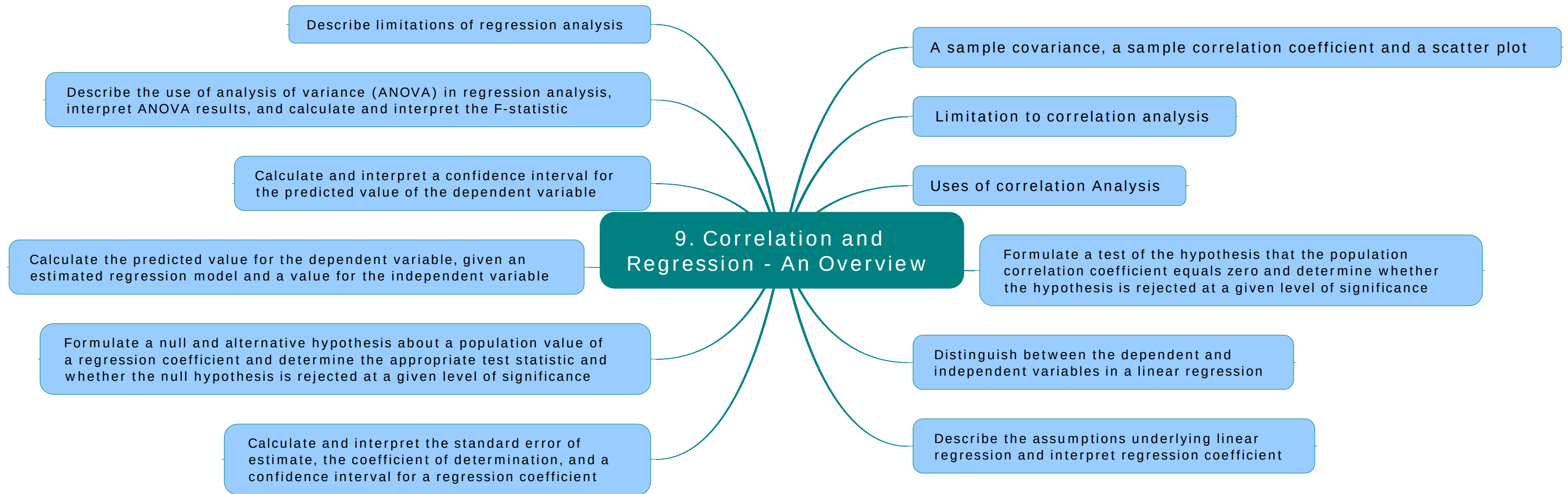
prohibit transactions done for tax purposes

The intent of action is critical to determining whether it is a violation of this Standard



To be continued...

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to:<http://www.e-junkie.com/ecom/gb.php?cl=274078&c=ib&aff=283565>



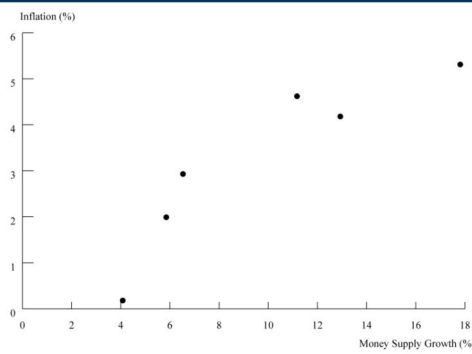
9. Correlation and Regression - Part 1

A sample covariance, a sample correlation coefficient and a scatter plot

Scatter Plots

A graph that shows the relationship between the observations for two data series in two dimensions

Figure 1. Scatter Plot of Annual Money Supply Growth Rate and Inflation Rate by Country, 1980-2012



Each observation in the scatter plot is represented as a point, and the points are not connected

The scatter shows only the actual observation of both data series plotted as pairs

Correlation Analysis

Correlation analysis expresses the same relationship (between two data series) using a single number

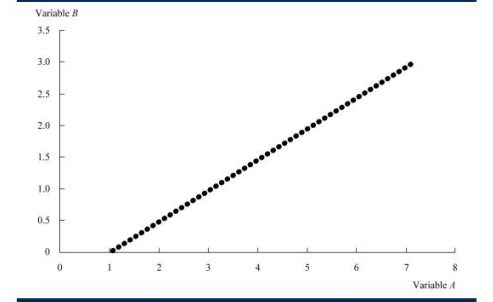
The correlation coefficient measures the direction and extent of linear association between two variables

A correlation coefficient less than 0 indicates a negative linear association

A correlation coefficient can have a maximum value of 1 and a minimum value of -1

A correlation coefficient greater than 0 indicates a positive linear association

Figure 2. Variables with a Correlation of 1



A scatter plot of two variables with a correlation of 0; they have no linear relation -> the value of A tells us nothing about the value of B

Calculate the Correlation Coefficient

The sample covariance of X and Y, for a sample of size n

$$\text{Cov}(X, Y) = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) / (n - 1)$$

The expression for the sample variance of X, is

$$s_X^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1) \text{ And } s_X = \sqrt{s_X^2}$$

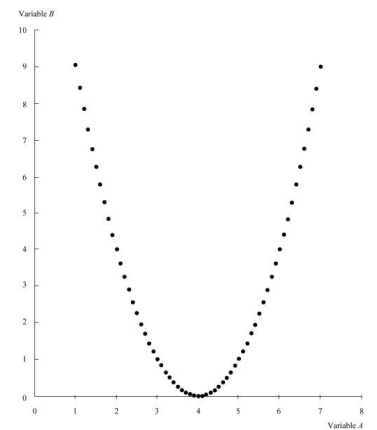
The sample correlation coefficient

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y}$$

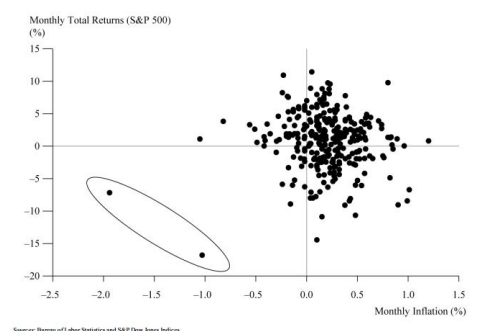
Limitation to correlation analysis

Correlation may be an unreliable measure when

Two variables can have a strong nonlinear relation and still have a very low correlation



Outliers are present in one or both of the series. Outliers are small numbers of observations at either extreme (small or large) of a sample



Spurious correlation

correlation between two variables that reflects chance relationship in a particular data set

correlation induced by a calculation that mixes each of two variables with a third

correlation between two variables arising not from a direct relation between them but from their relation to a third variable

Uses of correlation Analysis

In investment decision-making (for example: inflation forecast)

Correlation of stock market tells us how successfully the assets can be combined to diversify risk

Used in a financial statement setting

Formulate a test of the hypothesis that the population correlation coefficient equals zero and determine whether the hypothesis is rejected at a given level of significance

H0: the correlation in the population is 0 ($p = 0$)

Ha : the correlation in the population is different from 0 ($p \neq 0$)

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

The formula for the t-test

This test statistic has a t-distribution with n-2 degrees of freedom if the null hypothesis is true

Sampling from the same population, a false null hypothesis H0: is more likely to be rejected as we increase sample size. The result whether H0 is rejected also depends on significance level

Distinguish between the dependent and independent variables in a linear regression

Linear regression with one independent variable (or simple linear regression)

models the relationship between two variables as a straight line

Linear regression provides a simple model for forecasting the value of one variable, known as the dependent variable, given the value of the second variable, known as the independent variable

9. Correlation and Regression - Part 2

Describe the assumptions underlying linear regression and interpret regression coefficient

The regression equation

$$Y_i = b_0 + b_1 X_i + \varepsilon_i, i = 1, \dots, n$$

Y: dependent variable
X: independent variable
b0: the intercept
b1: a slope coefficient

b0, b1 are the regression coefficients

$$\hat{b}_1 = \frac{cov_{XY}}{\sigma_X^2}$$

Slope coefficient

The estimated slope coefficient is interpreted as the change in the dependent variable for a 1-unit change in the independent variable

The intercept term

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

$\bar{Y}$ = mean of Y

$\bar{X}$ = mean of X

The intercept is an estimate of the dependent variable when the independent variable takes on a value of zero

ε error term (represents the portion of the dependent variable that cannot be explained by the independent variable)

Six classic normal linear regression model assumptions

The relationship between the dependent variable, Y, and the independent variable, X is linear in the parameter b0 and b1. b0 and b1 are raised to the first power only and that neither b0 nor b1 is multiplied or divided by another (for example, b0/b1). The requirement does not exclude X from being raised to a power other than 1

Critical for a valid linear regression. If the relationship between the independent and dependent variables is nonlinear in the parameters, then estimating that relation with a linear regression model will produce invalid results

The independent variable, X, is not random

The expected value of the error term is 0 $E(\varepsilon) = 0$

Ensure that linear regression produces the correct estimates

The variance of the error term is the same for all observations: $E(\varepsilon_i^2) = \sigma_\varepsilon^2, i = 1, \dots, n$

The error term is uncorrelated across observations. Consequently, $E(\varepsilon_i, \varepsilon_j) = 0$ for all i not equal to j.

use the linear regression model to determine the distribution of the estimated parameters and and thus test whether those coefficients have a particular value

The error term is normally distributed

$$SEE = \left(\frac{\sum_{i=1}^n (Y_i - \hat{b}_0 - \hat{b}_1 X_i)^2}{n-2} \right)^{1/2} = \left(\frac{\sum_{i=1}^n (\hat{\varepsilon}_i)^2}{n-2} \right)^{1/2}$$

The formula for the standard error of estimate (SEE) for a linear regression model with one independent variable is

The different between the actual and predicted values of the dependent variable is the regression residual $\hat{\varepsilon}_i$

Calculate and interpret the standard error of estimate, the coefficient of determination, and a confidence interval for a regression coefficient

The coefficient of determination (R^2)

defined as the percentage of the total variation in the dependent variable explained by the independent variable

$R^2 = r^2$ for a regression with one independent variable

Regression coefficient confidence interval

Confidence interval spans the range

$$\hat{b}_1 - t_c s_{\hat{b}_1} \text{ to } \hat{b}_1 + t_c s_{\hat{b}_1}$$

A confidence interval is an interval of values that we believe includes the true parameter value, , with a given degree of confidence

Formulate a null and alternative hypothesis about a population value of a regression coefficient and determine the appropriate test statistic and whether the null hypothesis is rejected at a given level of significance

A hypothesis test using the confidence interval approach if we know

the estimated parameter value $\hat{b}_0$ or $\hat{b}_1$
the hypothesized value b0 or b1
a confidence interval around the estimated parameter

In practice, the most common way to test a hypothesis using a regression model is with a t-test of significance. To test the hypothesis, we can compute the statistic

$$t = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}}$$

This test statistic has a t-distribution with n-2 degrees of freedom. Reject H0 if t > +tcritical or t < -tcritical

The appropriate test structure for the null and alternative hypothesis: H0: b1 = 0 versus Ha: b1 # 0

Calculate the predicted value for the dependent variable, given an estimated regression model and a value for the independent variable

$$Y_i = b_0 + b_1 X_i + \varepsilon_i \quad \text{If we know } \hat{b}_0 \text{ or } \hat{b}_1 \quad \hat{Y} = \hat{b}_0 + \hat{b}_1 X$$

Calculate and interpret a confidence interval for the predicted value of the dependent variable

The prediction interval for a regression equation for a particular predicted value of the dependent variable Y $\hat{Y} \pm t_c s_f$ Where sf = standard error of the forecast
tc is two-tailed critical t-value at the desired level of significance with df = n-2

The formula to calculate sf
 $s_f^2 = SEE^2 \left[1 + \frac{1}{n} + \frac{(X - \bar{X})^2}{(n-1)s_x^2} \right]$
 SEE^2 variance of the residuals = the square of the standard error of estimate
 s_x^2 variance of the independent variable
X value of the independent variable for which the forecast was made

Describe the use of analysis of variance (ANOVA) in regression analysis, interpret ANOVA results, and calculate and interpret the F-statistic

Analysis of variance (ANOVA) is a statistical procedure for dividing the total variability of a variable into components that can be attributed to different sources
Use ANOVA to determine the usefulness of the independent variable or variables in explaining variation in the dependent variable
The F-test tests whether all the slope coefficients in a linear regression are equal to 0
The null hypothesis H0: b1 = 0
The alternative hypothesis Ha: b1 # 1

Formula for the F-statistic in a regression with one independent variable is

$$F = \frac{RSS/1}{SSE/(n-2)} = \frac{\text{Mean regression sum of squares}}{\text{Mean squared error}}$$

SSE (The sum of squared errors or residuals) $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

RSS (The regression sum of squares) $\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$

TSS = SSE + RSS

If there are n observations, the F-test for the null hypothesis that the slope coefficient is equal to 0 is denoted

$$F_{\# \text{ slope parameters}, n - \# \text{ parameters}} = F_{1, n-2}$$

$$R^2 = \frac{\text{total variation (TSS)} - \text{unexplained variation (SSE)}}{\text{total variation (TSS)}} = \frac{\text{explained variation (RSS)}}{\text{total variation (TSS)}}$$

$$SEE = \sqrt{MSE} = \sqrt{\frac{SSE}{n-2}}$$

Calculate R^2 and SEE

Describe limitations of regression analysis

Regression relations can change over time-> the issue of parameter instability
Public knowledge of regression relationships may negate their future usefulness
If the regression assumptions are violated, hypothesis tests and predictions based on linear regression will not be valid

29. Equity Valuation: Applications and Processes - An Overview

Introduction

Value Definition and
Valuation Applications

Communicating Valuation Results

The Valuation Process

Introduction

Valuation

Basic questions

The estimation of an asset's value based on variables perceived to be related to future investment returns, on comparisons with similar assets, or on estimates of immediate liquidation proceeds

- What is value?
- Who uses equity valuations?
- What is the importance of industry knowledge?
- How can the analyst effectively communicate his analysis?

The value of the asset given a hypothetically complete understanding of the asset's investment characteristics

Reflects investor's view of the "true" or "real" value of an asset

Market price and intrinsic value are identical

Grossman-Stiglitz paradox

Rational efficient markets formulation

Investors will not rationally incur the expenses of gathering information unless they expect to be rewarded by higher gross returns compared with the free alternative of accepting the market price

Common stock

Trading costs exist

Further room exists for price to diverge from value

Difficult to determine especially

Analysts often view market prices both with respect and with skepticism

Seek to identify mispricing

A difference between the estimated intrinsic value and the market price of an asset

Rely on price eventually converging to intrinsic value

Recognize distinctions among the levels of market efficiency or tiers of markets

Intrinsic Value

Valuation is an inherent part to attempt positive excess risk adjusted returns (abnormal return or alpha)

Uncertainty is constantly present

Reevaluate by looking for the presence of a particular market or corporate event (catalyst)

(VE-V): the difference between the valuation estimate and the true but unobservable intrinsic value

The error in the estimate of the intrinsic value

$$VE - P = (V - P) + (VE - V)$$

VE = estimated value

P = market price

V = intrinsic value

(V-P): the true mispricing, the difference between the true but unobservable intrinsic value V and the observed market price P

Contribute to the abnormal return

A useful estimate of intrinsic value

Combine accurate forecasts and appropriate valuation model

Active security selection

Manager's expectations must differ from consensus expectations and be correct

Expectational inputs used in valuation models

Going-Concern Value and Liquidation Value

Going-concern assumption

The assumption that the company will continue its business activities into the foreseeable future

value maximizing using assets

accessing its optimal sources of financing

not appropriate for a company in financial distress

The value added by assets working together and by human capital applied to managing those assets makes estimated goingconcern value greater than liquidation value

Liquidation value

Different time frame for liquidating causes different assets value of a company

Orderly liquidation value

Fair Market Value and Investment Value

Fair market value

is the price at which an asset (or liability) would change hands between a willing buyer and a willing seller when the former is not under any compulsion to buy and the latter is not under any compulsion to sell

includes an assumption that both buyer and seller are informed of all material aspects of the underlying investment

often used in valuation related to assessing taxes

Investment value

The concept of value to a specific buyer taking account of potential synergies and based on the investor's requirements and expectations

Valuation Applications

Selecting stocks

Primary use

Inferring (extracting) market expectations

evaluate the reasonableness of the expectations

as a benchmark or comparison value of the same characteristic for another company

Evaluating corporate events

A merger

the general term for the combination of two companies

An acquisition

a combination of two companies, with one of the companies identified as the acquirer, the other the acquired

the acquiring company's own common stock is often used as currency for the purchase

A divestiture

a company sells some major component of its business

A spin-off

the company separates one of its component businesses and transfers the ownership of the separated business to its shareholders

A leveraged buyout

an acquisition involving significant leverage (i.e., debt), which is often collateralized by the assets of the company being acquired.)

affects a company's future cash flows -> equity

Rendering fairness opinions

The parties to a merger may be required to seek a fairness opinion on the terms of the merger from a third party, such as an investment bank

Evaluating business strategies and models

Companies concerned with maximizing shareholder value evaluate the effect of alternative strategies on share value

Communicating with analysts and shareholders

Appraising private businesses

for transactional purposes

E.g acquisitions or buy-sell agreements for the transfer of equity interests among owners when one of them dies or retires, IPO,...

Sharebased payment (compensation)

Communicating Valuation Results

Contents of a Research Report

Kind of infor. intended readers seek to gain

Sell-side analyst's report: investment recommendation

Persuasive supporting arguments

The intrinsic value of the security

The key assumptions and expectations underlying that estimated intrinsic value

An update on the company's financial and operating results

A description of relevant aspects of the current macroeconomic and industry context

An analysis and forecast for the industry and company

Detailed historical descriptive statistics about the industry and company

May be accompanied by an explanation of the underlying rationale

Usual contents

Specific forecasts

A description of the valuation model

Key valuation inputs

A discussion of qualitative factors and other considerations that affect valuation

Objectively address the uncertainty associated with investing in the security, and/or the valuation inputs involving the greatest amount of uncertainty

An effective research report

Contains timely information

is written in clear, incisive language

is objective and well researched, with key assumptions clearly identified

distinguishes clearly between facts and opinions

contains analysis, forecasts, valuation, and a recommendation that are internally consistent

presents sufficient information to allow a reader to critique the valuation

states the key risk factors involved in an investment in the company

discloses any potential conflicts of interests faced by the analyst

The requirements are more specific in some situations. For e.g., regulations governing disclosures of conflicts and potential conflicts of interest vary across countries, investment recommendations are affected by policies of the firm employing an analyst

Section	Purpose	Content	Comments
Table of Contents	Show report's organization	Consistent with narrative in sequence and language	This is typically used in very long research reports only
Summary and Investment Conclusion	- Communicate the large picture - Communicate major specific conclusions of the analysis - Recommend an investment course of action	- Capable description of the company - Earnings projections - Other major conclusions - Valuation summary - Investment action	An executive summary; may be called simply "Summary."
Business Summary	- Present the company in more detail - Communicate a detailed understanding of the company's economics and current situation - Provide and explain specific forecasts	- Company description to the divisional level - Industry analysis - Competitive analysis - Historical performance - Financial forecasts	Reflects the first and second steps of the valuation process. Financial forecasts should be explained adequately and reflect quality of earnings analysis.
Risks	Alert readers to the risk factors in investing in the security	- Possible negative industry developments - Possible negative regulatory and legal developments - Possible negative company developments - Risks in the forecasts - Other risks	Readers should have enough information to determine how the analyst is defining and assessing the risks specific to investing in the security
Valuation	Communicate a clear and careful valuation	- Description of model(s) used - Recapitulation of inputs - Statement of conclusions	Readers should have enough information to critique the analysis
Historical and Pro Forma Tables	Organize and present data to support the analysis in the Business Summary		This is generally a separate section in longer research reports only. Many reports fold all or some of this information into the Business Summary section

Format of a Research Report

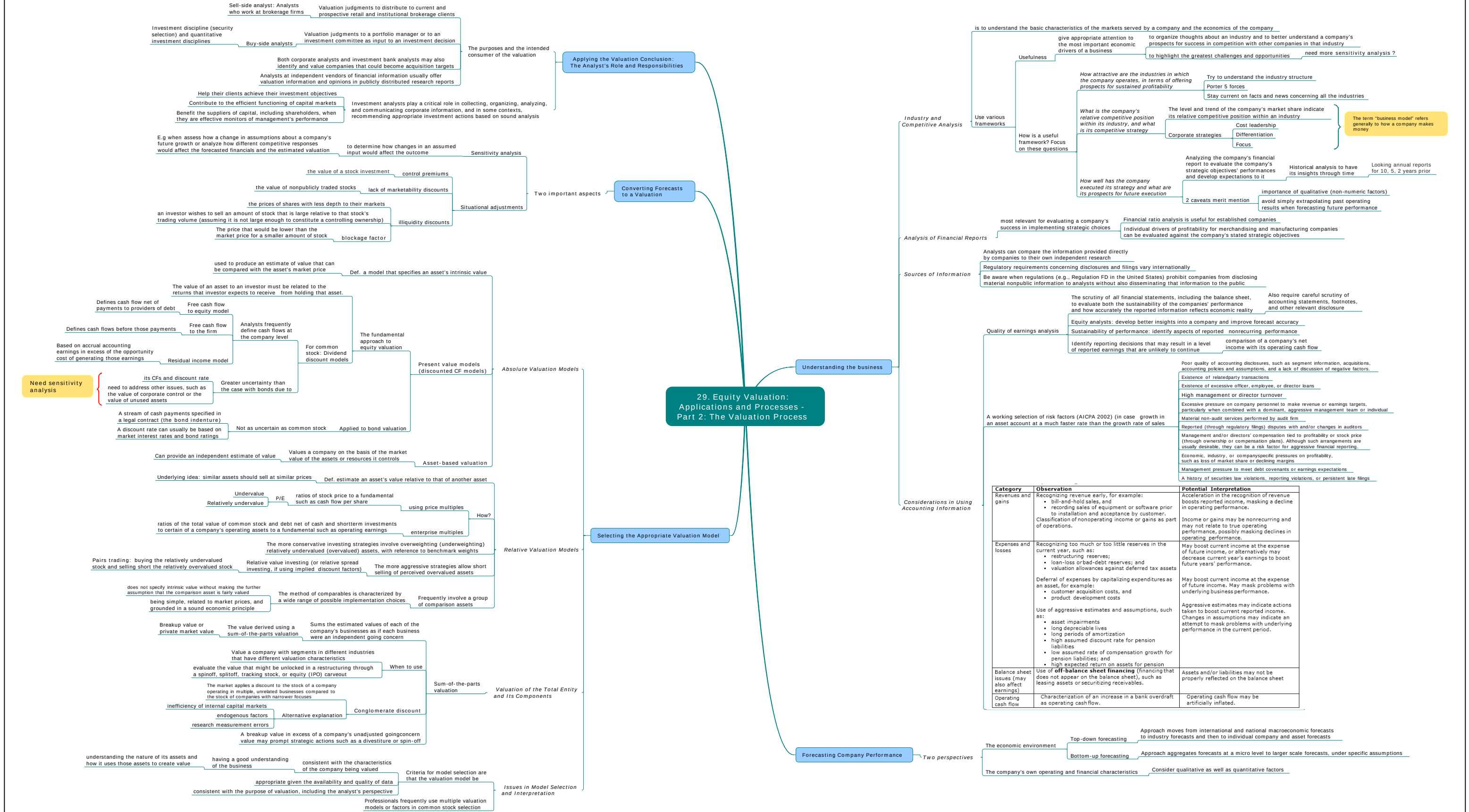
Research Reporting Responsibilities

The analyst must hold himself accountable to both standards of competence and standards of conduct

All analysts have an obligation to provide substantive and meaningful content in a clear and comprehensive report format

Analysts who are CFA Institute members, however, have an additional and overriding responsibility to adhere to the Code of Ethics and the Standards of Professional Conduct in all activities pertaining to their research reports

29. Equity Valuation: Applications and Processes - Part 1

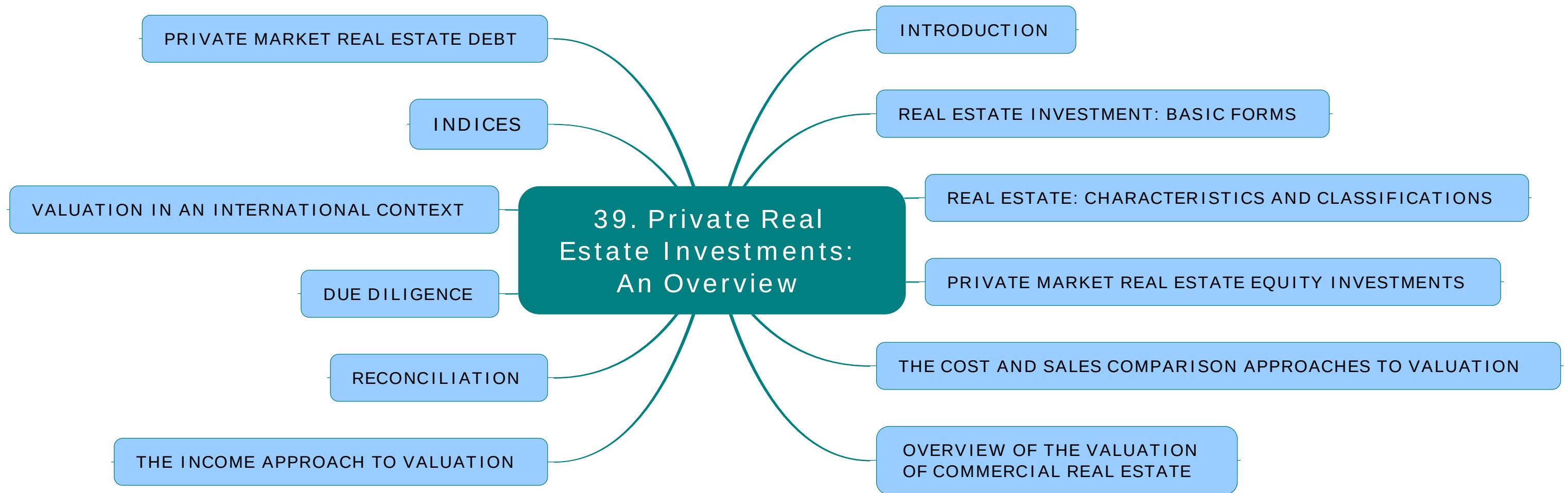




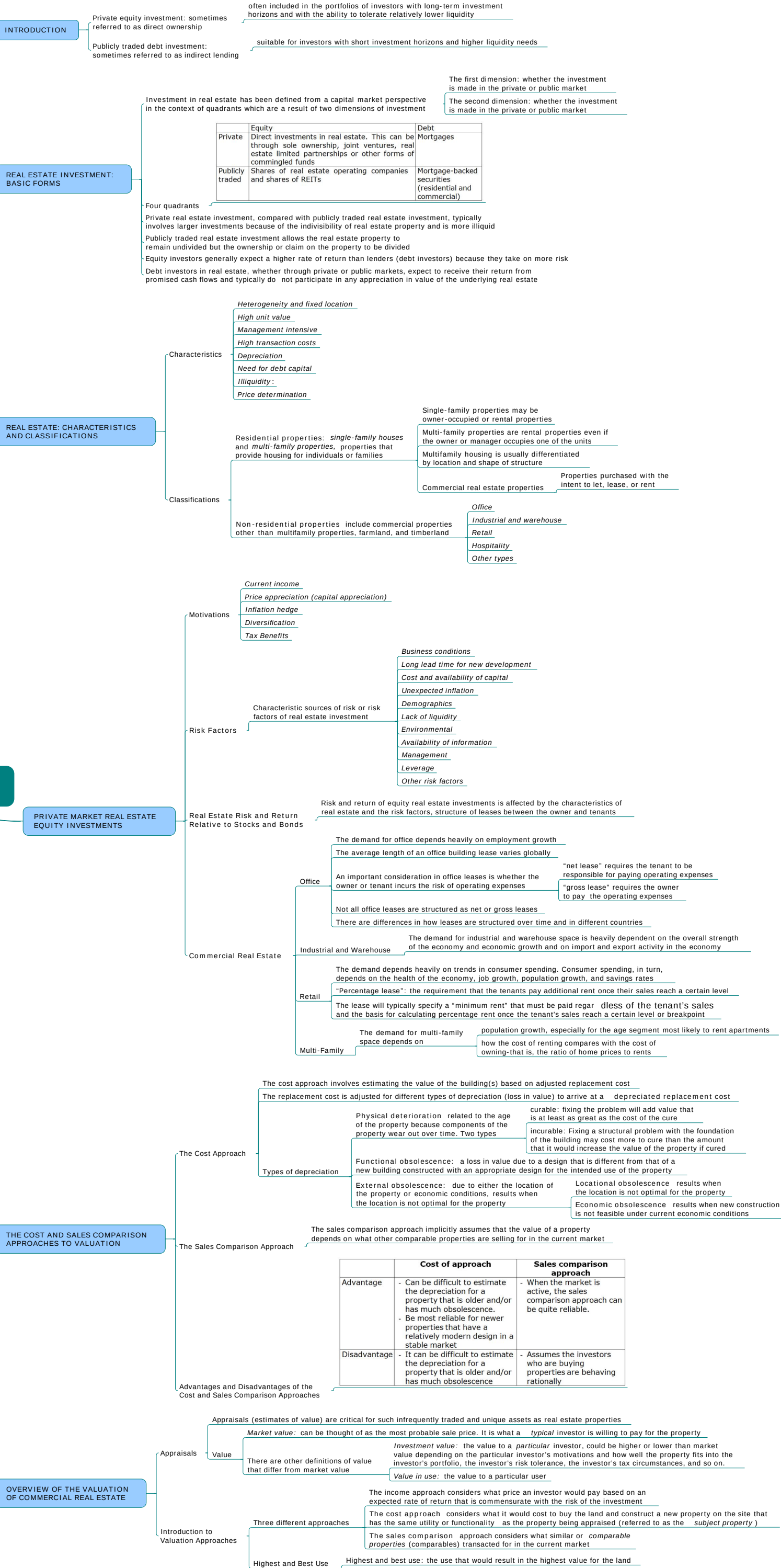
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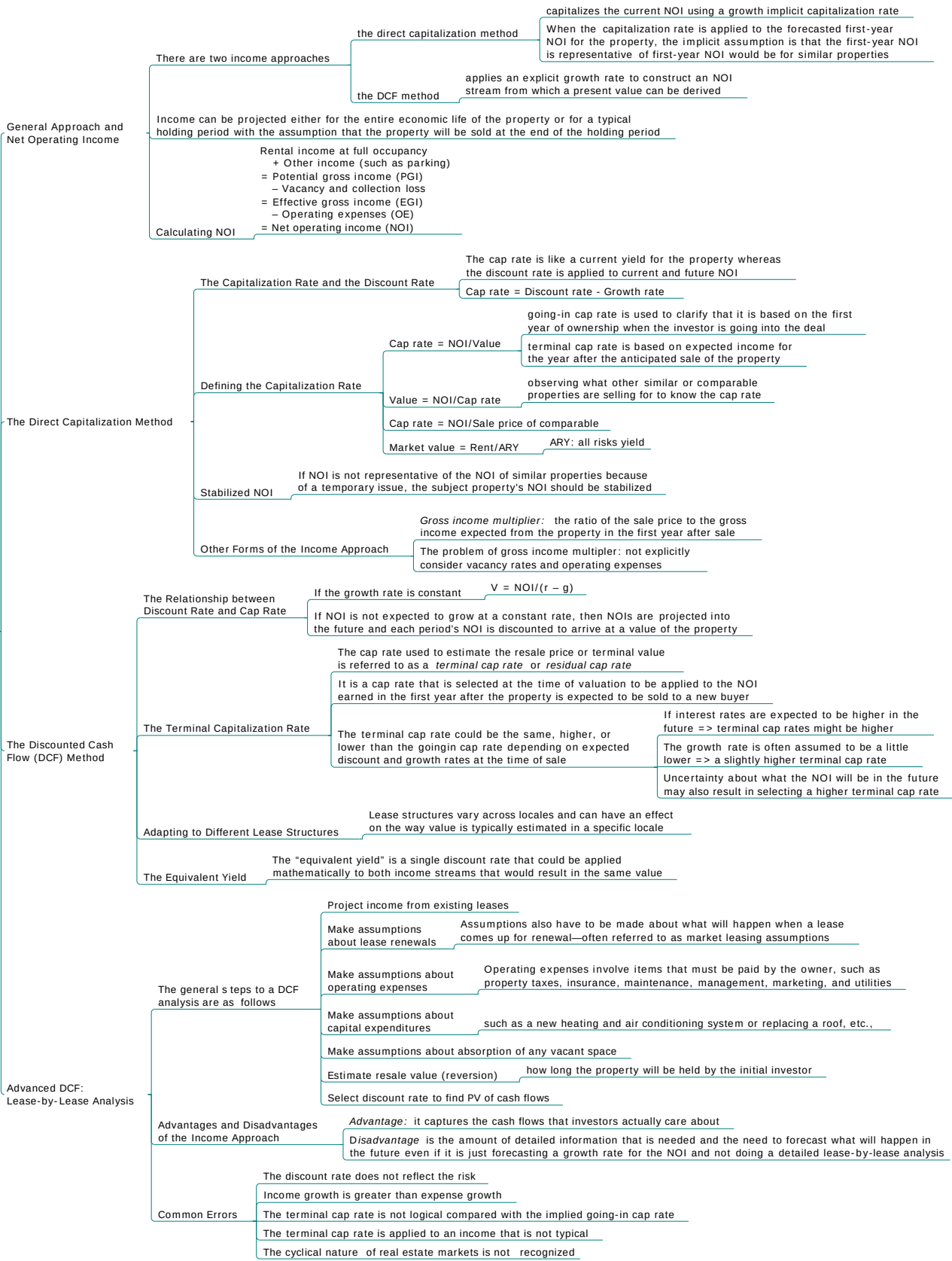


39. Private Real Estate Investments - Part 1



39. Private Real Estate Investments - Part 2

THE INCOME APPROACH TO VALUATION



RECONCILIATION

- Three different approaches to valuation: the income, cost, and sales comparison approaches may produce the different answers due to imperfections in the data and inefficiencies in the market
- The appraiser needs to *reconcile* the differences and arrive at a final conclusion about the value
- The purpose of reconciliation is to decide which approach or approaches you have the most confidence in and come up with a final estimate of value
- In an active market: sales comparison approach is preferred
- When there are fewer transactions: income approach is preferred

DUE DILIGENCE

- To verify other facts and conditions that might affect the value of the property and that might not have been identified by the appraiser
- E.g.
 - Review the leases for the major tenants and review the history of rental payments and any defaults or late payments.
 - Get copies of bills for operating expenses, such as utility expenses.
 - Look at cash flow statements of the previous owner for operating expenses and revenues.
 - Have an environmental inspection to be sure there are no issues, such as a contaminant material on the site.
 - Have a physical/engineering inspection to be sure there are no structural issues with the property and to check the condition of the building systems, structures, foundation, and adequacy of utilities.
 - Have an attorney or appropriate party review the ownership history to be sure there are no issues related to the seller's ability to transfer free and clear title that is not subject to any previously unidentified liens.
 - Review service and maintenance agreements to determine whether there are recurring problems.
 - Have a property survey to determine whether the physical improvements are in the boundary lines of the site and to find out if there are any easements that would affect the value.
 - Verify that the property is compliant with zoning, environmental regulations, parking ratios, and so on.
 - Verify that property taxes, insurance, special assessments, and so on, have been paid

VALUATION IN AN INTERNATIONAL CONTEXT

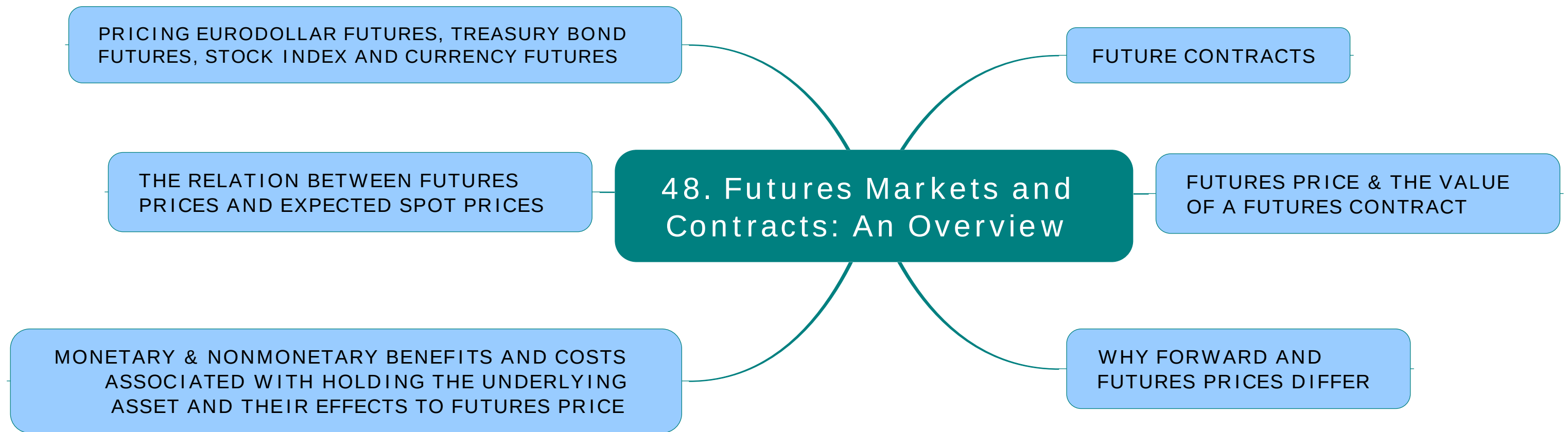
Summary of International Valuation Methodologies					
Country	Valuation Framework	Valuation Approaches	Lease Structure and Rent Reviews	Landlord vs. Tenant Expense Responsibility	Globalization
China	People's Republic of China and Regulations for Urban Land Valuation	Sales Comparison, Cost, and Income Approaches	2- to 3-year terms Upon expiry	Landlord: structure Tenant: interior, maintenance, insurance	DCF methodology gaining popularity
France	La Charte de l'Expertise en Evaluation Immobilière, (Le COB)	Sales Comparison and Cost Approaches, Comparative Implicit Capitalization	3-, 6-, 9-year terms Upon expiry, rental changes tie to INSEE Index	Landlord: structure Tenant: interior, maintenance, insurance	DCF gaining popularity with international investments/valuations
Germany	WertV, WertR, BelWertV	Sales Comparison, Cost, and Income Approaches	5-, 10-year terms Upon expiry, rental changes tie to cost of living index	Landlord: structure Dach und Fach Tenant: interior	DCF gaining popularity with international investments/valuations
Japan	Ministry of Land, Infrastructure, Transport and Tourism	Sales Comparison, Cost, and Income Approaches	3-, 5-year terms 6 months prior to lease expiration	Landlord: repair Tenant: inside maintenance, insurance	DCF used as an analysis tool and widely used in international investments/valuations
United Kingdom	Royal Institution of Chartered Surveyors (RICS) Red Book	Sales Comparison and Cost Approaches, Implicit Capitalization	Recently changed from 25 years to 10, 15 years 5-year upward only	Landlord: minimal Tenant: all repairs, insurance	DCF used as an analysis tool and widely used in international investments/valuations
United States	Universal Standards of Professional Appraisal Practice (USPAP)	Sales Comparison and Cost Approaches, Implicit Capitalization, DCF	3-, 5-year terms	10+ years Upon expiry	Depends on lease structure

INDICES

- Appraisal-Based Indices
 - Return = $\{NOI - \text{Capital expenditures} + (\text{Ending market value} - \text{Beginning market value})\} / \text{Beginning market value}$
 - Disadvantages
 - Appraisal lag
 - May not capture the price increase until a quarter or more after it was reflected in transactions
 - Tend to smooth the index, have lower correlation with others => allocation to real estate would likely overestimated
 - How to adjust: unsmooth" the appraisalbased or use a transactionbased index when comparing real estate with other asset classes
- Transaction-Based Indices
 - In recent years, indices have been created that are based on actual transactions rather than appraised values
 - Two main ways
 - A repeat sales index relies on repeat sales of the same property
 - A hedonic index which requires only one sale
 - Disadvantages
 - Include random elements in the observations => may be upward or downward movements from quarter to quarter that are somewhat random

PRIVATE MARKET REAL ESTATE DEBT

- The maximum amount of debt that an investor can obtain on commercial real estate is usually limited by either the ratio of the loan to the appraised value of the property (loan to value or LTV) or the debt service coverage ratio (DSCR)
- The debt service coverage ratio is the ratio of the firstyear NOI to the loan payment
 - DSCR = NOI/Debt service



48. Futures Markets and Contracts - Part 1

FUTURE CONTRACTS

- Similar to forward contracts in
- Different from forward contracts

Deliverable contracts obligate the long to buy and the short to sell a certain quantity of an asset for a certain price on a specified future date

Cash settlement contracts are settled by paying the contract value in cash on the expiration date

Both forwards and futures are priced to have zero value at the time the investor enters into the contract

Futures are marked to market at the end of every trading day. Forward contracts are not marked to market

Futures contracts trade on organized exchanges. Forwards are private contracts and do not trade on organized exchanges

Futures contracts are highly standardized. Forwards are customized contracts satisfying the needs of the parties involved

Forwards are contracts with the originating counterparty; a specialized entity called a clearinghouse is the counterparty to all futures contracts

Forward contracts are usually not regulated. The government having legal jurisdiction regulates futures markets

FUTURES PRICE & THE VALUE OF A FUTURES CONTRACT

- Futures price must converge to the spot price at expiration
- Value of a futures contract

At expiration, the spot price must equal the futures price because the futures price has become the price today for delivery today, which is the same as the spot.

Arbitrage will force the prices to be the same at contract expiration

The clearinghouse guarantees that traders in the futures market will honor their obligations by splitting each trade once it is made and acting as the opposite side of each position => To safeguard the clearinghouse, both sides of the trade are required to post margin and settle their accounts on a daily basis

Futures margin is a performance guarantee

Marking to market is the process of adjusting the margin balance in a futures account each day for the change in the value of the contract from the previous trading day, based on the settlement price

Has no value at contract initiation

Does not accumulate value changes over the term of the contract.

The value after the margin deposit has been adjusted for the day's gains and losses in contract value is always zero

The futures price at any point in time is the price that makes the value of a new contract equal to zero

The value of a futures contract strays from zero only during the trading periods between the times at which the account is marked to market

Value of futures contract = current futures price - previous mark-to-market price

If the futures price increases, the value of the long position increases

$$FP = S_0 \times (1 + R_f)^T$$

FP = futures price
S₀ = spot price at inception of the contract (t = 0)
R_f = annual risk-free rate
T = futures contract term in years

should be the same as that of a forward contract

The no-arbitrage price of a futures contract

Cases that causes futures and forward prices to be different

If investors prefer the mark-to-market feature of futures, futures prices will be higher than forward prices

If investors would rather hold a forward contract to avoid the marking to market of a futures contract, the forward price would be higher than the futures price

WHY FORWARD AND FUTURES PRICES DIFFER

<i>If the correlation between the underlying asset value and interest rates is ...</i>	<i>Investors will ...</i>
Positive	Prefer to go long in a futures contract, and the futures price will be greater than the price of an otherwise comparable forward contract
Zero	Have no preference
Negative	Prefer to go long in a forward contract, and the forward price will be greater than the price of an otherwise comparable futures contract

Cash-and-carry arbitrage

A cash-and-carry arbitrage consists of buying the asset, storing/holding the asset, and selling the asset at the futures price when the contract expires

Steps

- At the initiation of the contract
 - Borrow money for the term of the contract at market interest rates
 - Buy the underlying asset at the spot price
 - Sell (go short) a futures contract at the current futures price
- At contract expiration
 - Deliver the asset and receive the futures contract price
 - Repay the loan plus interest

If the futures contract is overpriced => generate a riskless profit

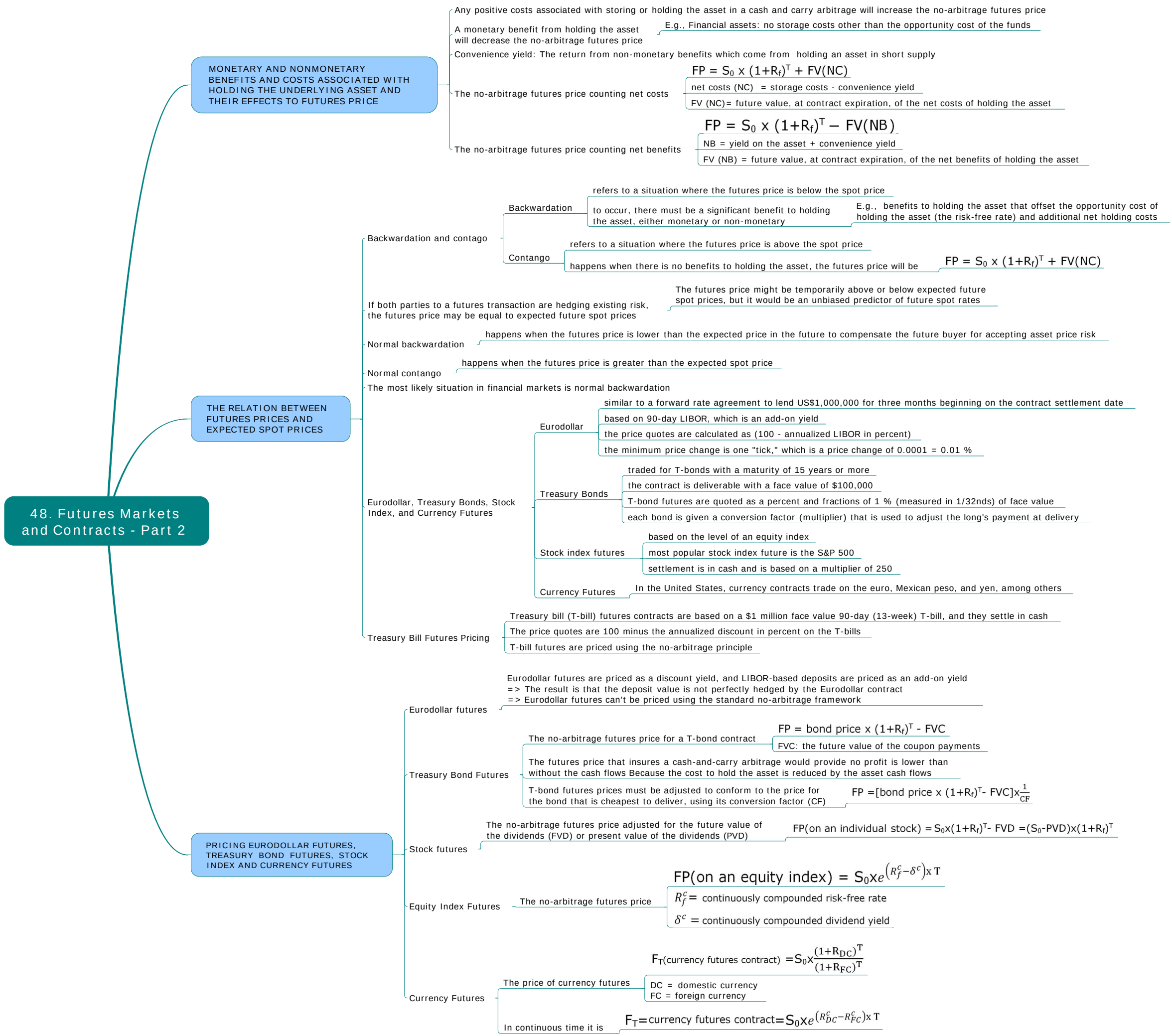
The futures contract is overpriced if the actual market price is greater than the no-arbitrage price

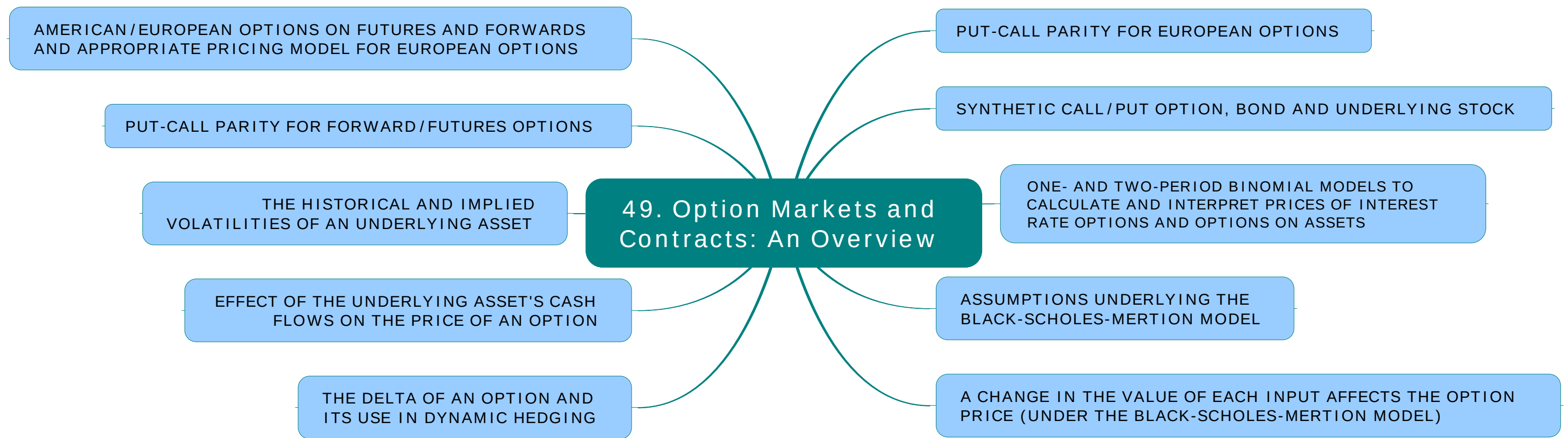
Reverse cash-and-carry arbitrage

When the futures price is too low (which presents a profitable arbitrage opportunity)

Steps

- At the initiation of the contract
 - Sell the asset short
 - Lend the short sale proceeds at market interest rates
 - Buy (go long) the futures contract at the market price
- At contract expiration
 - Collect the loan proceeds
 - Take delivery of the asset for the futures price and cover the short sale commitment





49. Option Markets and Contracts - Part 1

PUT-CALL PARITY FOR EUROPEAN OPTIONS

A fiduciary call

A long position in a European call option with an exercise price of X that matures in T years on a stock (with a price at time t of S_t)

A long position in a pure-discount riskless bond that pays X in T years

Payoff to a Fiduciary Call

	$S_T \leq X$ Call is out-of or at-the-money	$S_T > X$ Call is in-the-money
Long call payoff	0	$S_T - X$
Long bond payoff	X	X
Total payoff	X	S_T

A protective put

A long position in a European put option with an exercise price of X that matures in T years

A long position in the underlying stock

Payoff to a Protective Put

	$S_T < X$ Put is in-the-money	$S_T \geq X$ Put is out-of or at-the-money
Long put payoff	$X - S_T$	0
Long stock payoff	S_T	S_T
Total payoff	X	S_T

That the cost of a fiduciary call must be equal to the cost of a protective up

$$C_0 + \left[\frac{X}{(1+R_f)^T} \right] = P_0 + S_0$$

+ : long position
- : short position

Put-call parity for European options

A synthetic European call option is formed by

Buying a European put option on the same stock with the same exercise price (X) and the same maturity (T)

Buying the stock

Shorting (i.e., borrowing) the present value of X worth of a pure-discount riskless bond

A synthetic European put option is formed by

Buying a European call option

Shorting the stock

Buying (i.e., investing in) the discount bond

A synthetic stock position is formed by

Buying a European call option

Shorting (i.e., writing) a European put option

Buying (i.e., investing in) the discount bond

A synthetic pure-discount riskless bond is created by

Buying a European put option

Buying the stock

Shorting (i.e., writing) a European call option

Two reasons to create synthetic positions in the securities

To price options by using combinations of other instruments with known prices

To earn arbitrage profits by exploiting relative mispricing among the four securities

Using put-call parity for arbitrage

If put-call parity doesn't hold (if the cost of a fiduciary call does not equal the cost of a protective put), buy (go long in) the underpriced position and sell (go short) in the overpriced position

ONE- AND TWO-PERIOD BINOMIAL MODELS TO CALCULATE AND INTERPRET PRICES OF INTEREST RATE OPTIONS AND OPTIONS ON ASSETS

One-Period Binomial Model

$$\Pi_u = \text{risk-neutral probability of an up-move} = \frac{1+R_f-D}{U-D}$$

$$\Pi_D = \text{risk-neutral probability of a down-move} = 1 - \Pi_u$$

R_f = risk-free rate
 U = size of an up-move
 D = size of a down-move

Calculate the value of an option on the stock

Calculating the payoff of the option at maturity in both the up-move and down-move states

Calculating the expected value of the option in one year as the probability-weighted average of the payoffs in each state

Discounting the expected value back to today at the risk-free rate

Arbitrage with one-period binomial model

$$\Delta = \frac{C_1^+ - C_1^-}{S_1^+ - S_1^-}$$

provides the information required to calculate a hedge ratio (the fractional share of stock in the arbitrage trade)

Two-Period Binomial Model

Steps to value an option

Calculate the stock values at the end of two periods (there are three possible outcomes because an up-then-down move gets you to the same place as a down-then-up move)

Calculate three possible option payoffs at the end of two periods

Calculate the expected option values at the end of two periods ($t = 2$) using the up-and down-move probabilities

Discount the expected option values ($t = 2$) back one period at the risk-free rate to find the option values at the end of the first period ($t = 1$)

Calculate the expected option value at the end of one period ($t = 1$) using up-and down-move probabilities

Discount the expected option value at the end of one period ($t = 1$) back one period at the risk-free rate to find the option value today

Binomial interest rate trees

is the set of possible interest rate paths that are used to value bonds with a binomial model

the underlying rule governing the construction of an interest rate tree

the values for on-the-run issues generated using an interest rate tree should prohibit arbitrage opportunities

the interest rate tree must maintain the interest rate volatility assumption of the underlying model

Options on Fixed Income Securities

There are three basic steps to valuing an option on a fixed-income instrument using a binomial tree

price the bond at each node using projected interest rates

calculate the intrinsic value of the option at each node at maturity of the option, and

calculate the value of the option today

Options on Interest Rates: Caps and Floors

The value of an interest rate cap or floor is the sum of the values of the individual caplets or floorlets

$$\text{Expiration value of caplet} = \frac{\max \{0, [(one\text{-}year\ rate - cap\ rate) \times notional\ principal]\}}{1 + one\text{-}year\ rate}$$

$$\text{Expiration value of floorlet} = \frac{\max \{0, [(floor\ rate - one\text{-}year\ rate) \times notional\ principal]\}}{1 + one\text{-}year\ rate}$$

As the period covered by a binomial model is divided into arbitrarily small, discrete time periods, the model results converge to those of the continuous-time model

The Black-Scholes-Merton (BSM) model values options in continuous time and is derived from the same no-arbitrage assumption used to value options with the binomial model

To derive the BSM model, an "instantaneously" riskless portfolio is used to solve for the option price based on the same logic

The price of the underlying asset follows a lognormal distribution

The (continuous) risk-free rate is constant and known

Limitation: The BSM model is not useful for pricing options on bond prices and interest rates

The volatility of the underlying asset is constant and known

In practice, the volatility is not known and must be estimated. The bigger problem is that volatility is often not constant over time and the BSM model is not useful in these situations

Markets are "frictionless"

Model is less realistic and less useful

The underlying asset generates no cash flows

The BSM model can be easily altered if we relax the assumption of no cash flows on the underlying asset

The options are European

The model does not correctly price American options. Binomial option pricing models are more appropriate for pricing American options

ASSUMPTIONS UNDERLYING THE BLACK-SCHOLES-MERTON MODEL

The formula for the BSM model

$$C_0 = [S_0 \times N(d_1)] - [X \times e^{-R_f^c \times T} \times N(d_2)]$$

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + [R_f^c + (0.5 \times \sigma^2)] \times T}{\sigma \times \sqrt{T}}$$

$$d_2 = d_1 - (\sigma \times \sqrt{T})$$

T = time to maturity (as % of a 365-day year)

S_0 = asset price

X = exercise price

R_f^c = continuously compounded risk-free rate

σ = volatility of continuously compounded returns on the stock

$N()$ = cumulative normal probability of value in parentheses

Use put-call parity to calculate the put value

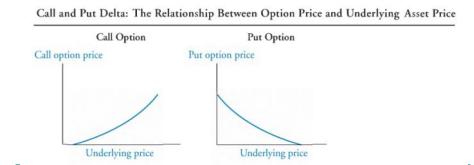
$$P_0 = C_0 - S_0 + (X \times e^{-R_f^c \times T})$$

49. Option Markets and Contracts - Part 2

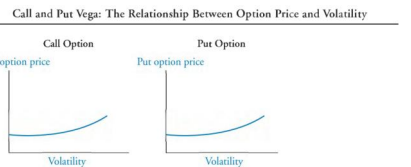
A CHANGE IN THE VALUE OF EACH INPUT AFFECTS THE OPTION PRICE (UNDER THE BLACK-SCHOLES-MERTION MODEL)

A Greek is a sensitivity factor that captures the relationship between each input (asset price, asset price volatility, time to expiration, and the risk-free rate) the option price

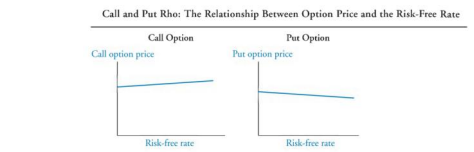
Delta describes the relationship between asset price and option price



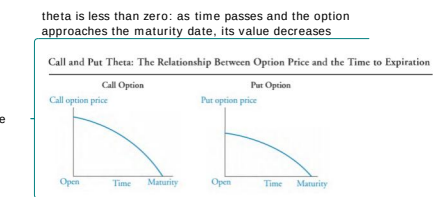
Vega measures the sensitivity of the option price to changes in the volatility of returns on the underlying asset



Rho measures the sensitivity of the option price to changes in the risk-free rate



Theta measures the sensitivity of the option price to the passage of time



Direction of BSM European Option Prices for a Change in the Five Model Inputs

Sensitivity Factor ("Greek")	Input	Calls	Puts
Delta	Asset price (\$)	Positively related Delta > 0	Negatively related Delta < 0
Vega	Volatility (σ)	Positively related Vega > 0	Positively related Vega > 0
Rho	Risk-free rate (r)	Positively related Rho > 0	Negatively related Rho < 0
Theta	Time to expiration (T)	Value $\rightarrow$ \$0 as call $\rightarrow$ maturity Theta < 0	Value usually $\rightarrow$ \$0 as put $\rightarrow$ maturity Theta < 0*
	Exercise price (X)	Negatively related	Positively related

AMERICAN / EUROPEAN OPTIONS ON FUTURES AND FORWARDS AND APPROPRIATE PRICING MODEL FOR EUROPEAN OPTIONS

There is a benefit to early exercise of options on futures when they are deep in the money
Exercising the option (either a put or call) early will generate cash from the mark to market => cash can earn interest, while the futures position will gain or lose from movements in the futures price => these price movements between early exercise and option expiration will mirror those of the deep in the money option
There is no mark to market on forwards, early exercise does not accelerate the payment of any gains
With no reason for early exercise, the value of American and European options on forwards are the same
American options on futures are more valuable than comparable European options because

$$C_0 = e^{-R_f^f T} [F_T \times N(d_1) - X \times N(d_2)]$$
$$d_1 = \frac{\ln\left(\frac{F_T}{X}\right) + \left(\frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}$$
$$d_2 = d_1 - \sigma \sqrt{T}$$

The Black model can be used to price European options on forwards and futures

σ = standard deviation of returns on the futures contract
 F_T = futures price

PUT-CALL PARITY FOR FORWARD / FUTURES OPTIONS

$$C_0 + \frac{X - F_T}{(1 + R_f)^T} = P_0$$

American options on futures are more valuable than European options because early exercise provides mark to market funds on the futures, which can earn interest
Americans and European options on forward contracts are equivalent because there is no mark to market

THE HISTORICAL AND IMPLIED VOLATILITIES OF AN UNDERLYING ASSET

$$R_i = \frac{P_i - P_{i-1}}{P_{i-1}}, i = 1 \text{ to } N$$

Step 1: Convert a time series of N prices to returns

$$R_i^c = \ln(1 + R_i), i = 1 \text{ to } N$$

Step 2: Convert the returns to continuously compounded returns

$$\sigma^2 = \frac{\sum_{i=1}^N (R_i^c - \bar{R}_i^c)^2}{N - 1}$$

$$\sigma = \sqrt{\sigma^2}$$

Step 3: Calculate the variance and standard deviation of the continuously compounded returns

The steps in computing historical volatility for use as an input in the BSM continuous-time options pricing model are

when used in the Black-Scholes formula, it produces the current market price of the option
Implied volatility is the value for standard deviation of continuously compounded rates of return that is "implied" by the market price of the option

EFFECT OF THE UNDERLYING ASSET'S CASH FLOWS ON THE PRICE OF AN OPTION

Decrease the value of a call option
Increase the value of a put option

All else equal, the existence of cash flows on the underlying asset will

$$C_0 + \frac{X}{(1 + R_f)^T} = P_0 + (S_0 - PVCF)$$

Put-call parity for options on underlying assets with cash flows by adjusting S for the present value of the cash flows (PVCF)

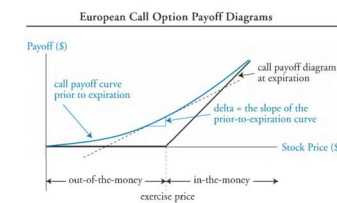
THE DELTA OF AN OPTION AND ITS USE IN DYNAMIC HEDGING

Delta is the change in the price of an option for a one-unit change in the price of the underlying security

$$\text{Delta}_{\text{call}} = \frac{C_1 - C_0}{S_1 - S_0} = \frac{\Delta C}{\Delta S}$$

ΔC = change in the price of the call over a short time interval
 ΔS = change in the price of the underlying stock over a short time interval

Use BSM model to estimate the change in the value of the call given the change in the value of the stock and the option's delta
 $\Delta C \approx N(d_1) \times \Delta S$
 ΔP (change in put price) $\approx [N(d_1) - 1] \times \Delta S$



Interpreting Delta

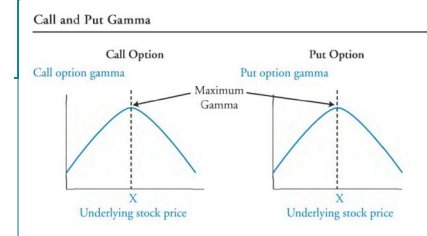
A call option delta is between 0 and 1. If the call option is
Out-of-the-money (stock price is less than exercise price), the call delta moves closer to 0 as time passes, assuming the underlying stock price doesn't change
In-the-money (stock price is greater than exercise price), the call delta moves closer to 1 as time passes, assuming the underlying stock price doesn't change
A put option delta is between -1 and 0. If the put option is
Out-of-the-money (stock price is greater than exercise price), the put delta moves closer to 0 as time passes, assuming the underlying stock price doesn't change
In-the-money (stock price is less than exercise price), the put delta moves closer to -1 as time passes, assuming the underlying stock price doesn't change

Dynamic Hedging

The goal of a delta-neutral portfolio (or delta-neutral hedge) is to combine a long position in a stock with a short position in a call option so that the value of the portfolio does not change when the value of the stock changes
Number of call options needed to delta hedge = number of shares hedged/delta of call option
Number of put options needed to delta hedge = number of shares/delta of the put option
The delta-neutral position only holds for very small changes in the value of the underlying stock => must be continually rebalanced to maintain the hedge (a dynamic hedge) costly in terms of transaction costs

Gamma measures the rate of change in delta as the underlying stock price changes
can be viewed as a measure of how poorly a dynamic hedge will perform when it is not rebalanced in response to a change in the asset price

Call and put options on the same underlying asset with the same exercise price and time to maturity will have equal gammas
Long positions in calls and puts have positive gammas
Gamma is largest when a call or put option is at-the-money and close to expiration





To be continued...

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